

LAK 2026 · WORKSHOP

The Utility of Synthetic Data

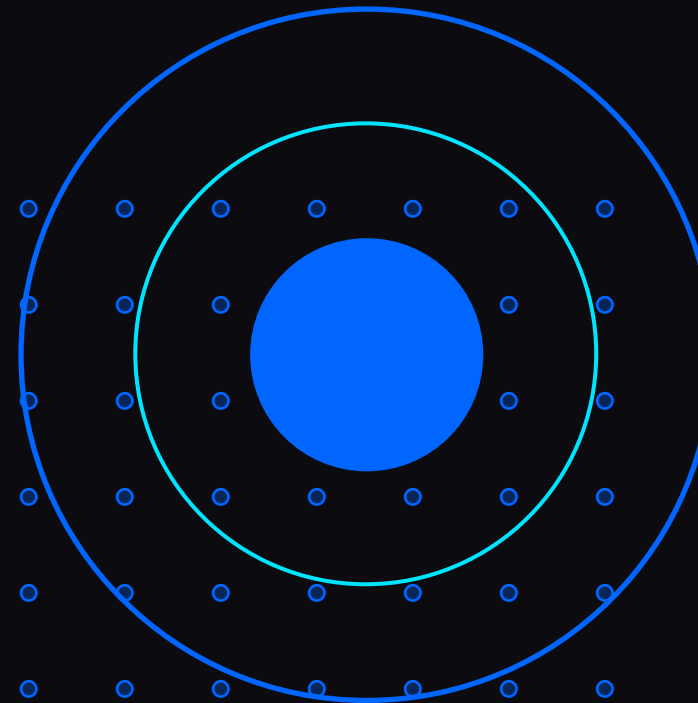
Why it matters for learners, teachers, and researchers

Sam Urmian



CENTRE FOR THE SCIENCE OF LEARNING & TECHNOLOGY (SLATE)

A workshop on synthetic data for education research



lasd.slaterresearch.ai



The education data paradox

We collect more student data than ever. Most of it never reaches the people who could learn from it.

< 5%

of educational datasets are openly shareable across institutions

Privacy reviews, FERPA, GDPR, IRB — each a months-long handshake.



Privacy walls

Real student records carry FERPA, GDPR, and IRB constraints. Sharing across labs is rare.



Sparse populations

Rare students — under-represented groups, struggling learners — vanish in small samples.



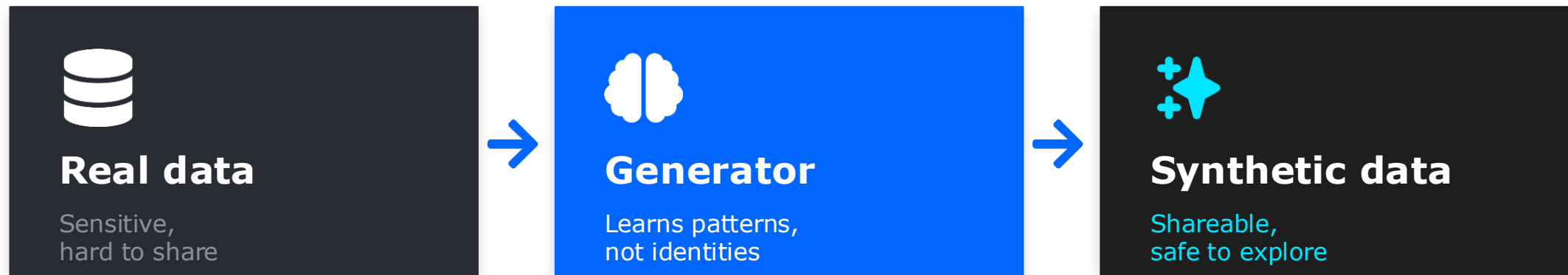
Reproducibility gap

Findings can't be replicated when the underlying data can't leave its institution.



So, what is synthetic data?

Data that **looks and behaves like real student data** — but belongs to **no real student**.





Real vs synthetic — at a glance



REAL DATA

Source	Actual student records
Privacy	Restricted — IRB & consent required
Volume	Whatever the institution collected
Sharing	Months of approvals, often blocked
Bias	The truth, including the ugly parts



SYNTHETIC DATA

Source	Generated from a learned model
Privacy	No real identities — much lower risk
Volume	As much as you need
Sharing	Drop into a paper, repo, or workshop
Bias	Whatever the source data — and the model — encoded



How it's made — three flavours

All three start with real data. They differ in how clever the recipe is.

01



Statistical sampling

Fit a distribution to each variable, then draw new rows from it.

Simple. Fast. Loses the joins.

02



Generative ML

Train a GAN, VAE, or diffusion model to mimic the joint structure.

Captures correlations. Needs tuning.

03



LLM-driven

Prompt a language model to invent realistic learners and trajectories.

Rich, narrative-friendly. Hard to validate.



Where utility lives — four pillars

Synthetic data is useful when it unlocks one of these. If it doesn't, don't bother.



Privacy

Share data without sharing students.
Lower disclosure risk by design.



Access

Open the data tap for collaborators,
classrooms, and reviewers.



Fairness

Stress-test models on populations the real
data under-samples.



Innovation

Prototype, teach, and benchmark long
before the IRB letter arrives.



Sharing data without sharing students

A learning analytics dataset travels — no student does.



STORY

From the field

Your district collected three years of LMS clickstream data. A research team in another country wants to study procrastination patterns. Real records can't cross the border; a synthetic copy can.



What happens

A model is trained on the real logs and never leaves the district. Only the model — or its outputs — is shared.



Why it works

Patterns travel. Identities stay home. The receiving team works with realistic data on day one.



Watch out for

Memorisation of rare cases. Use disclosure-risk audits and differential-privacy training where possible.



Filling the gaps for rare learners

Augment under-represented populations so they're not invisible.



STORY

From the field

A reading-intervention study has plenty of typical learners — but only 14 students with a specific learning disability. Models trained on this set will silently overlook them.



What happens

Targeted synthetic samples thicken the tails of the distribution where the real data is thin.



Why it works

Downstream models train on a more representative population — without inventing students from nothing.



Watch out for

Augmenting from a tiny seed risks amplifying its quirks. Anchor with domain expertise.



Stress-testing for fairness

What happens when the algorithm meets a learner it rarely sees?



STORY

From the field

An automated grading model performs well on average — but does it perform well for English-language learners? For students who type slowly? For students whose answers are unusually long?



What happens

Synthetic learners are crafted along controlled axes — language background, response length, learning trajectory.



Why it works

You can probe failure modes you'd never have enough real data to test cleanly.



Watch out for

Synthetic doesn't mean neutral. The probe inherits the assumptions of whoever built it.



Common ground for the field

Open benchmarks turn private work into shared progress.



STORY

From the field

Two papers claim the same modelling technique works — on two private datasets nobody else can touch. Replication, comparison, and incremental progress all stall.



What happens

A community releases a synthetic benchmark that mirrors the structure of real datasets it can't release.



Why it works

Methods are compared on the same terms. New researchers can join without an institutional handshake.



Watch out for

Benchmarks can drift from reality. Treat them as a floor, not a finish line.



A safe sandbox for students

Teach learning analytics with data that looks like the real thing.



STORY

From the field

Your graduate seminar needs realistic LMS data to practise on. The real records are governed by an IRB protocol that doesn't list 32 first-year masters students.



What happens

Students work with synthetic records that mirror real schemas, scales, and edge cases.



Why it works

Skills transfer to real datasets — with none of the consent paperwork or breach risk in the classroom.



Watch out for

Be explicit that conclusions drawn from synthetic data are about the data, not the students.



Where the utility shows up

A quick map of the wins — most projects cash in on two or three at once.



Lower privacy risk

No real identities to leak.



Faster collaboration

Skip months of data agreements.



Richer populations

Boost rare and minority groups.



Fairness checks

Probe edge cases on demand.



Reproducible work

Benchmarks anyone can rerun.



Hands-on teaching

Realistic, classroom-safe data.



Faster prototyping

Build before the IRB lands.



Inclusive research

Smaller institutions get a seat.



Honest about the trade-offs

Synthetic data is a powerful tool — not a silver bullet. The honest list:



Bias travels. If the source data under-represents a group, the synthetic copy will too — sometimes more so.



Validation is hard. There's no universal test for 'realistic enough'. Utility depends on what you're going to do next.



Privacy is not automatic. Models can memorise. Without controls — like differential privacy — rare records can leak through.



Distribution shift sneaks in. Synthetic data is frozen in the moment its source was. Real classrooms keep moving.



It's not a substitute for consent. Synthetic data complements an ethics culture; it doesn't replace one.



Reach for it when... and skip it when...

REACH FOR IT WHEN

-  Privacy stops a real-data project before it starts.
-  Rare or under-represented learners need a fairer test.
-  You're prototyping, teaching, or sharing a benchmark.
-  You need data right now and the IRB letter is six months out.
-  You want collaborators outside your institution to play along.

THINK TWICE WHEN

-  Decisions about real students hinge on the result.
-  The source data is already too small to learn a model from.
-  You can't validate the synthetic against any held-out reality.
-  It would be used to bypass consent or oversight conversations.
-  Stakes are high and the model's failure modes aren't audited.

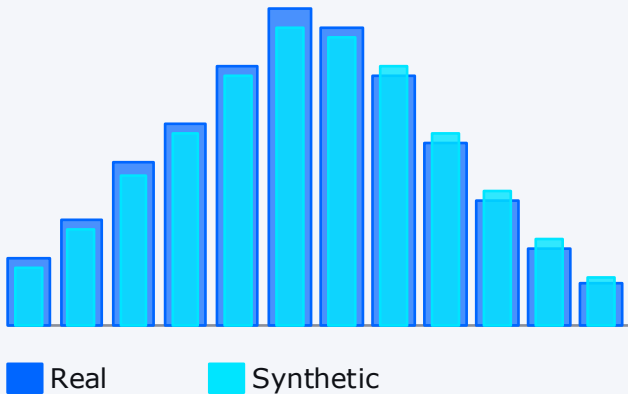


Fidelity — does it look like the real thing?

Statistical resemblance between real and synthetic data, on three levels.

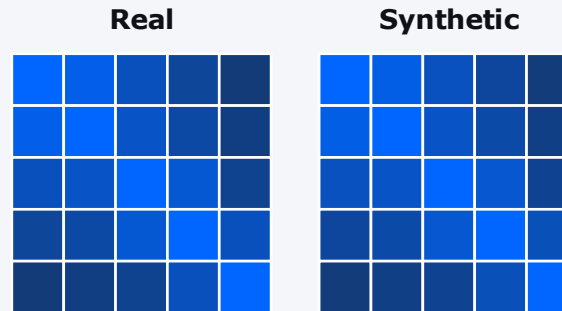
MARGINALS

Per-column distributions overlap.



CORRELATIONS

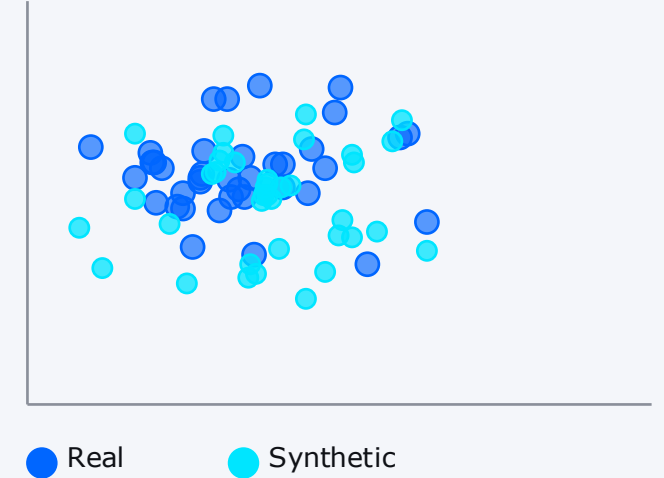
Pairwise structure preserved.



low ○ high correlation

PROJECTIONS

Clouds occupy the same space.





Fidelity $\neq$ Utility

Looking realistic isn't enough — synthetic data has to work for the task.



Beautiful histograms, broken model

A model can mimic every marginal yet destroy the joint structure that downstream tasks rely on.



Sometimes 'wrong' is right

Slightly off distributions can still train an excellent classifier — if the task-relevant signal is preserved.



Always test the use case

Pick a downstream evaluation that mirrors what you'd actually do with real data.



SPOTLIGHT · OPEN-SOURCE LIBRARY

TabDM · diffusion for mixed-type tabular data

A small Python library for generating synthetic tabular data with diffusion in a transformed feature space — built with evaluation baked in.

WHAT'S IN THE BOX



Mixed types

numeric, categorical, boolean, ordinal, count, positive continuous



Conditional

fix targets, balance subgroups, control sensitive columns



Built-in eval

schema, distribution (TVD/KS), validity, downstream utility, trust



Privacy screen

exact-match rate and nearest-neighbor diagnostics



python

```
from tabdm import generate_synthetic_data

synthetic = generate_synthetic_data(
    real,
    target_column="at_risk",
    sensitive_columns=["gender"],
    num_rows=10_000,
)
```

github.com/mlgorithm/tabdm · `pip install tabdm` · Sam Urmian & Mohammad Khalil · University of Bergen

From walled gardens to open research.



Synthetic data won't replace real students.
But it can let many more people learn from how they learn.



Workshop

Join the LAK 2026 discussion



Materials

lasd.slaterresearch.ai



Bring back

One use case to your work